PRANJAL KARAN

Computer Science Engineering Student

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Education

VIT Bhopal University – Bachelor of Technology, Computer Science & Engineering — Sehore, MP **Sept 2023 – May 2027**

CGPA: 8.70/10

Technical Skills

Languages: JavaScript (ES6+), C++, SQL

Backend & APIs: Node.js, Express.js, RESTful API Design, WebSockets, JWT Authentication, Clerk Auth

Databases: MongoDB (Mongoose, Aggregation Pipelines, Compound & Partial Unique Indexes), MySQL, PostgreSQL

Frontend: React, HTML5, CSS3, Tailwind CSS v4, Vite, Responsive Web Design

AI & Emerging Tech: GitHub Copilot, Claude, ChatGPT, Antigravity, LLM API Integration (Gemini, Ollama)

Tools & Practices: Git, GitHub, Vercel, Docker, ESP8266, Blynk IoT, SDLC, Agile Problem-Solving

Soft Skills: Cross-Functional Collaboration, Adaptability, Team Leadership

Projects

HealthPulse – Full-Stack Healthcare Appointment & Clinical Platform — Live — GitHub **2026** React 18, TypeScript, Node.js, Express.js, MongoDB, JWT, Google Gemini, Ollama

- **Architected** a **40,000+ line** full-stack platform with **61 REST API endpoints** across 9 modules and **13 MongoDB schemas**, following a strict controller-service-model architecture.
- **Eliminated** double-booking race conditions under concurrent load by combining a **MongoDB compound partial unique index** with **TTL-based 5-minute slot holds**, guaranteeing atomic booking integrity at the database engine tier.
- **Engineered** stateless JWT authentication with a `tokenVersion` mechanism enabling instant session revocation, role-based access control across 3 personas, and rate-limited endpoints to prevent brute-force attacks.
- **Orchestrated** a dual-engine AI clinical-summary pipeline (Google Gemini + Ollama via `Promise.allSettled`) with a zero-hallucination medication-verification guard, backed by **154 automated tests (471 assertions)** and a **98/100** Lighthouse performance score.

VibeChat – Real-Time Full-Stack Messaging Platform — GitHub **2026** React 19, Tailwind v4, Zustand, Node.js, MongoDB, Socket.io, Clerk, ImageKit, Docker

- **Engineered** a real-time messaging application achieving **sub-100ms** message distribution using custom Socket.io server channels, replacing N+1 queries with a **6-stage Mongoose aggregation pipeline**.
- **Packaged** a lightweight **1.17MB** production bundle via Vite 8 featuring Clerk OAuth session gates and ImageKit lazy-loading pipelines.

AstroSync – AI-Powered Compatibility Analysis Platform — Live — GitHub **2026** React, TypeScript, Vite, Vercel Serverless Functions, Google Gemini API

- **Built** a TypeScript single-page application with 10 algorithmic scoring modules and interactive Recharts-based visualizations for compatibility analysis.
- **Developed** 2 Vercel serverless functions integrating the Google Gemini API with multi-model fallback and chat-history normalization to handle context-window and rate-limit constraints.

Achievements & Certifications

Achievements:

- **National Qualifier** – Cleared Round 1 of **TCS CodeVita**, a national-level competitive coding challenge
- **Co-Lead (Social Media)**, Linpack Club, VIT Bhopal – Increased event engagement by **30%** through targeted digital campaigns

Certifications:

- **Microsoft Certified: Azure AI Cloud Developer Associate** (Score: 922/1000 — ID: DDFC0151CEFD415C)
- **Microsoft Certified: Azure Data Fundamentals** (Score: 955/1000)
- **MERN Full Stack Certification** (Ethnus & Codemithra)